

B.I.T. SINDRI, DHANBAD
MECHANICAL ENGINEERING DEPARTMENT

1st MID SEMESTER EXAMINATION-2024
B. TECH. FIRST SEMESTER
BRANCH- MECH., CIVIL, MET., CHEM., MIN., PROD.

SUBJECT- EDCG

TIME - $1\frac{1}{2}$ Hrs.

FULL MARKS : 10

Instructions :-

- The questions are of equal values.
- Assume any missing data or values suitably.
- Draw neat and labeled Drawing.
- Use H and 2H pencils only.
- Student can use both sides of Drawing sheet
- Keep proper dimensioning.
- Attempt any two questions.

1. Draw the projections of the following points on the same ground line keeping the projectors 30 mm apart. Proper dimensioning must be done as per BIS standards. [CO1]

- (i) Point A is 50 mm above the HP and 30 mm in front of the VP.
- (ii) Point B is 40 mm above the HP and on the VP.
- (iii) Point C is 30 mm in front of the VP and on the HP.
- (iv) Point D is 30 mm above the HP and 50 mm behind the VP.
- (v) Point E is on the HP and 40 mm behind the VP.
- (vi) Point F is 25 mm below the HP and 45 mm behind the VP.
- (vii) Point G is 30 mm below the HP and 45 mm in front of the V.P.
- (viii) Point H is on the V.P. and 30 mm below the H.P.
- (ix) Point I is 30 mm below the H.P. and 50 mm behind the V.P.
- (x) Point J is both on the H.P. and the V.P.

2. The front view and top view of a straight line PQ measures 50 mm and 65 mm respectively. Point p is in the H.P. and 20 mm in front of the V.P. and the front view of the line is inclined at 45° to the reference line. Determine the true of PQ, true angles of inclination with the reference planes and trace. [CO1]

3. A line AB, 90 mm long is inclined at 30° to the H.P. Its end A is 12 mm above H.P. and 20 mm in front of the V.P. Its front view measures 65 mm. Draw the top view of AB and determine its inclination with the V.P. Also, locate the V.T. and H.T. of the line. [CO1]

B.TECH., FIRST SEMESTER
BRANCHES: ME, PE, ML, CL, CE AND MN
SECOND MID SEMESTER EXAMINATION, 2025
ENGINEERING DRAWING AND COMPUTER GRAPHICS (EDCG)

Full Marks: 20

Time: 1 Hour 30 Minutes

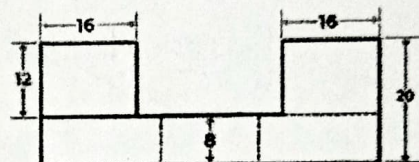
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2	4	0	5	1	3	2
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Important Instructions:

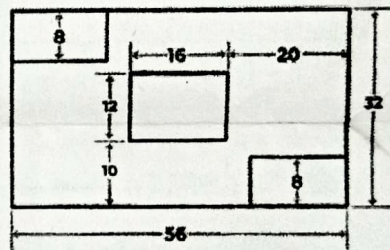
1. ALL THE QUESTIONS CARRY EQUAL MARKS.
2. ANSWER ANY TWO QUESTIONS.
3. The full marks corresponding to a question is indicated in right braces.
4. Assume any missing data or values suitably.
5. Write your Roll No. on the question paper in the space provided.
6. Do not write anything else on your question paper.
7. Use of a non-programmable calculator is permitted.
8. Use of any Log Tables, Slide Rules, Data Books etc. is not permitted.

1. The orthographic projections of a three-dimensional object is shown in Figure 1. Draw its isometric projection. [10]



Front View

2. The isometric projection of a three-dimensional object is shown in Figure 2. Draw its orthographic projection (Plan, Elevation and Left-Side View). [10]



Top View

Figure 1 (Question No. 1)

3. A pictorial view of a machine bracket is shown in the Figure 3. Draw its Plan, Elevation and Right-Side View. [10]

4. A hexagonal pyramid, base side 25 mm and axis 80 mm long, has its base horizontal and one of the edges of the base parallel to the V.P. A horizontal section plane cuts it at a distance 40 mm above the base. Draw the isometric projection of the frustum of the pyramid. [10]

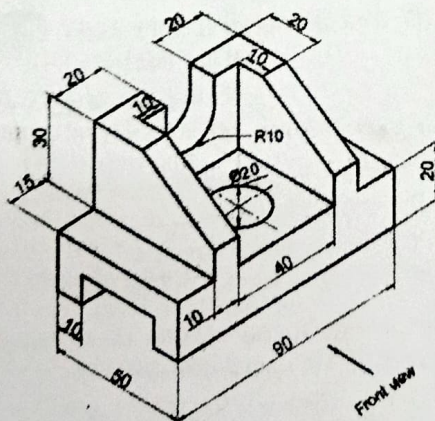


Figure 2 (Question No. 2)

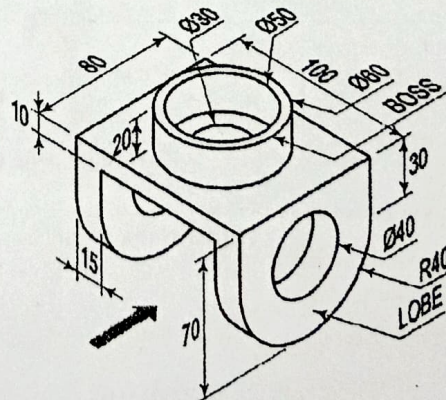


Figure 3 (Question No. 3)

All dimensions in figures are in mm.

5. The top view of a 75 mm long line AB measures 65 mm, while the length of its front view is 50 mm. It's one end A is in the H.P. and 12 mm in front of the V.P. Draw the projections of line AB and determine its inclinations with the H.P. and the V.P. [10]

THE END

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MECH/CIVIL/METAL/CHEM/PROD/MINING/FT

Jharkhand University of Technology, Ranchi

B.Tech. 1st Semester Examination, 2023 (NEP)

Subject : Engineering Drawing & Computer Graphics

Subject Code : ESED2

Time Allowed : 3 Hours

Full Marks : 70

Candidates are requested to give the answers in their own words as far as practicable.

Answer any five questions in which Question No. 1 is compulsory.

All questions are of equal value.

Assume any missing data or values suitably.

1. Answer the following questions :

2×7=14

(i) A point P is 25 mm below the HP and 25 mm in front of the Vp. State the quadrant in which the point is situated.

- (a) Point P is situated in the 2nd quadrant. (b) Point P is situated in the 4th quadrant.
(c) Point P is situated in the 1st quadrant. (d) None of these

(ii) If a line is parallel to a reference plane, it means that both of its ends are _____ from that plane.

- (a) non-equidistant (b) equidistant
(c) Both (a) and (b) (d) None of these

(iii) _____ is defined as the intersection of the line on the reference plane when it is moved towards it without changing its direction.

- (a) Trace of the line (b) True length of the line
(c) Apparent length of the line (d) None of these

(iv) When a line is inclined to both VP and HP but parallel to PP (Profile Plane) then sum of its inclination will be

- (a) $\theta + \phi = 45^\circ$ (b) $\theta + \phi = 90^\circ$
(c) $\alpha + \beta = 45^\circ$ (d) $\theta + \phi = 120^\circ$

(v) Isometric projection of a sphere is draw with true (length) radius. But its centre is located with _____.

- (a) isometric height (b) true height
(c) half of the true height (d) half of the isometric height

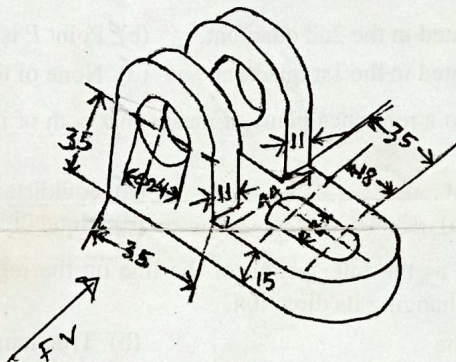
(vi) In an orthographic projection approaches, if a face is perpendicular to the direction of viewing, its _____ will be seen in that view.

- (a) tilted shape and size (b) apparent shape and size
(c) true shape and size (d) comparatively smaller shape and size

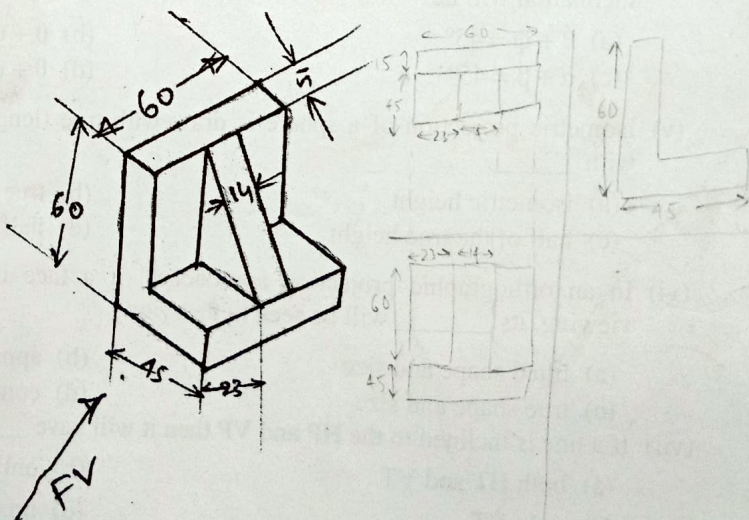
(vii) If a line is inclined to the HP and VP then it will have _____.

- (a) both HT and VT (b) only HT
(c) only VT (d) no traces in this case

2. A line AB, 50 mm long, has its end A away from the HP and the VP than end B. The line is inclined to the HP at 30° and to the VP at 45° . Draw the projections if end A is 35 mm above the HP and 50 mm in front of the VP. 14
3. Line CD is in the second quadrant and has 25° inclination with HP, while the front view has 30° inclination with xy line and 60 mm length. If the end C is 12 mm above HP and the end D is 60 mm behind VP, draw its projections. 14
4. A cylindrical block of base, 60 mm diameter and height 90 mm, standing on the HP with its axis perpendicular to the HP. Draw its isometric view. 14
5. A pentagonal pyramid of base side 30 mm and axis length 65 mm is resting on HP on its base with a side of base perpendicular to VP. It is cut by a plane inclined at 30° to HP and perpendicular to VP and passes through a point at a distance 30 mm from the apex. Draw the Isometric projection of the truncated pyramid. 14
6. The pictorial view of bearing block is shown below. Draw the front view, top view and side view according to the 1st angle projection method. (All dimensions are in mm.) 14



7. The pictorial view of a block is shown below (Fig. is not to scale). Draw the top view, front view and side view according to the first angle projection method. All dimensions are in mm. 14



ED2

MECH/CIVIL/METAL/CHEM/PROD/MINING/FT

Jharkhand University of Technology, Ranchi
B.Tech. 1st Semester Examination, 2024 (NEP)

Subject : Engineering Drawing & Computer Graphics

Subject Code : ESED2

Time Allowed : 3 Hours

Full Marks : 70

The questions are of equal values.

Answer/Attempt any five questions (Including Q.N.-1).

Question No. 1 is compulsory.

Assume any missing data or values suitably.

1. Choose correct option.

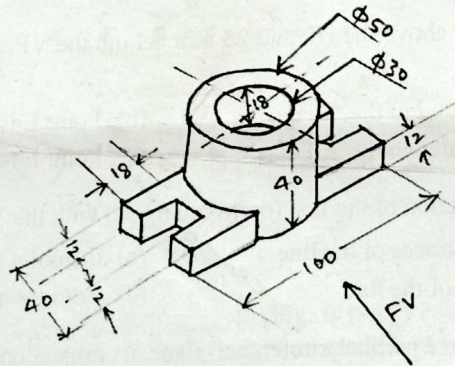
7×2=14

- (i) If a point moves in a plane, then the locus obtained is a _____.
(a) space curve (b) line curve
(c) 3D curve (d) None of these
- (ii) A point E is 20 mm above the HP and 25 mm behind the VP. State the quadrant in which the point is located.
(a) Point E is situated in the 4th quadrant. (b) Point E is situated in the 2nd quadrant.
(c) Point E is situated in the 3rd quadrant. (d) Point E is situated in the 1st quadrant.
- (iii) The point of intersection of the line (or its extension) with the VP is called _____.
(a) the horizontal traces of the line (b) the point view of the line
(c) vertical traces of the line (d) Both (a) and (c)
- (iv) If a line is parallel to a particular reference plane, its projection on that reference plane gives
(a) apparent length (b) line view with larger in length
(c) true length (d) line view with smaller in length
- (v) The three lines meeting at a point and making 120° angles with each other are termed _____.
(a) isometric axes (b) isometric planes
(c) non-isometric lines (d) non-isometric planes
- (vi) Isometric projection of an object is seen _____ in size than its actual size.
(a) bigger (b) smaller
(c) equal (d) 50% smaller
- (vii) In an orthographic projection approaches, if a face is perpendicular to the direction of viewing, its _____ will be seen in that view.
(a) tilted shape and size (b) comparatively smaller shape and size
(c) true shape and size (d) apparent shape and size

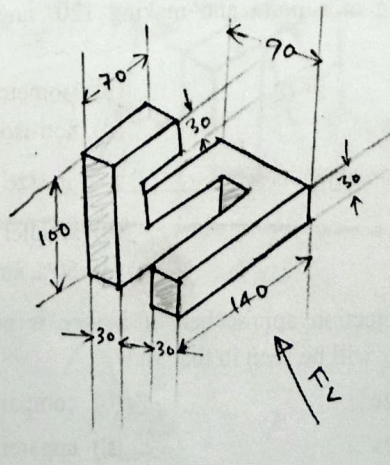
ED2

(2)

2. The TV of a line CD measures 80 mm and makes an angle of 55° with XY. End C is in the VP and the HT of the line is 25 mm above XY. The line is inclined at 30° to the HP. Draw the projections of line CD. Determine its True Length (TL), true inclination with the VP and the VT. 14
3. The front view of a line PQ makes an angle of 30° with XY. The HT of the line is 45 mm behind the VP. While its VT is 30 mm above the HP. The end P of the line is 10 mm below the HP and the end Q is in the first quadrant. The line is 150 mm long. Draw the projections of the line and determine the true length of the portion of the line which is in the second quadrant. Also find the angle of the line with the HP and VP. 14
4. A cylindrical block of base, 60 mm diameter and height 90 mm, standing on the HP with its axis perpendicular to the HP. Draw its isometric view. 14
5. A hexagonal prism of side of base 20 mm and length 50 mm is resting on one of its rectangular faces upon HP, keeping the base parallel to VP. The prism is cut by a vertical section plane 30° inclined to VP and passing through the mid-point on the axis. Draw isometric view of the prism showing the sectional surface. 14
6. The pictorial view of a cylindrical block is shown below (Figure is not to scale). Draw the top view, front view and side view according to the first angle projection method. (All dimensions are in mm.) 14



7. Figure shows an object formed of square sectioned steel bar. Draw its three principal views. (All dimensions are in mm.) 14



BIT Sindri
Department of Chemistry
B. Tech 1st Semester
(For Mech, Civil, Metal, Chemical, Production and Mining Branches)
2nd Mid-Semester Examination-2024
(Batch 2024-28)

Subject: Engineering Chemistry

Time : 1 hr. 30Min

Subject Code : BSC 02

Full Marks : 10

Candidates are required to give their answers in their own words as far as practicable.

Answer any five questions in which Question No.1 is compulsory

Q. No	Questions	Full Marks	COs
Q.1	All the Questions are multiple choices with one correct answer. Write the correct answer. (A) A type of electrical battery which can be charged, discharged into load, and then recharge many times is called----- (a) Rechargeable battery (b) Secondary battery (c) Primary battery (d) Both (a) and (b) (B) Which element is used for making solar cell? (a) Gold (b) Iron (c) Aluminium (d) Silicon (C) A Fuel cell is a -- (a) Galvanic cell (b) Electrolytic cell (c) Electrolytic Concentration Cells (d) None of these (D) What is the typical size range of nanoparticle? (a) 10-100 micrometers (b) 1-10 micrometers (c) 100-1000 nanometers (d) 1-100 nanometers	$\frac{1}{2} \times 4 = 2$	CO3 & CO4
Q.2	(A) Explain the Fundamental Principle of Lithium-ion battery, Including the role of anode, cathode, electrolyte and separator. (B) How does a sodium -ion battery differ from lithium-ion battery	1+1 =2	CO3
Q.3	(A) Write Working Principle and Name the different types of Fuel Cell. (B) Write Working Principle and Name the different types of solar Cell.	1+1 =2	CO3
Q.4	(A) What is green hydrogen technology? Discuss its applications. (B) Write difference between monocrystalline solar panels with polycrystalline solar panels.	1+1 =2	CO3
Q.5	(A) Define Nanomaterials. How can you classify into different categories on the basis of number of dimensions? (B) Write Applications of Nanomaterials in Electronic and Electrical Industries.	1+1 =2	CO3
Q.6	(A) Define Super capacitors. Discuss the classification of Supercapacitors. (B) Discuss RF Sputtering deposition method to synthesize nanomaterials. Give application of this methods.	1+1 =2	CO4
Q.7	Write notes on any Four of the following: (a) Quantum dots (b) Synthesis of a Super capacitors (c) Thin layer deposition method (d) Alkali Fuel cell (e) Principle of Green Chemistry	$\frac{1}{2} \times 4 = 2$	CO3 & CO4

BIT Sindri
Department of Chemistry
B. Tech 1st Semester *FOR NON - Ckt. BRANCH*
1st Mid-Semester Examination-2024
(Batch 2024-28)

Subject: Engineering Chemistry

Time : 1 hr. 30Min

Subject Code : BSC 02

Full Marks : 10

Candidates are required to give their answers in their own words as far as practicable.

Answer any five questions in which Question No.1 is compulsory

Q. No	Questions	Full Marks	COs
Q.1	All the Questions are multiple choices with one correct answer. Write the correct answer. (A) An important technique employed for the separation and purification of compound is -- (a) Chromatography (b) Conductometry (c) Potentiometry (d) Coulometry (B) The Mode of excitation of molecule in IR spectroscopy is - (a) Electronic (b) Vibrational (c) Magnetic (d) none of these (C) Which of the following is an example of QA? (a) Verification (b) Documentation (c) Both (d) none of these (D) The metal which is protected by its own oxide (a) Cu (b) Fe (c) Au (d) Al	$\frac{1}{2} \times 4 = 2$	CO1 & CO2
Q.2	(A) Define and explain Quality Control and Quality Assurance in Chemical Analysis. (B) Write difference between Qualitative and Quantitative analysis in Analytical Chemistry	1+1=2	CO1
Q.3	(A) How do you classify materials for Engineering use? (B) Describe in detail the role of Engineering Materials in our present world.	1+1=2	CO1
Q.4	(A) Define Infrared spectroscopy. Describe the various molecular vibrations in this technique. (B) Define and Explain Bathochromic Shift in UV spectroscopy	1+1=2	CO1
Q.5	(A) What are Laws of Dry Corrosion? Explain with suitable examples. (B) Discuss in details the various methods of corrosion Control with chemical equations	1+1=2	CO2
Q.6	(A) Discuss the factors that enhance rate of Corrosion. (B) Differentiate between wet and dry Corrosion.	1+1=2	CO2
Q.7	Write notes on any Four of the following: (a) Waterline Corrosion (b) Cathodic protection of metal (c) Mechanical properties of an Engineering material (d) Thermogravimetry Analysis (e) Corrosion Inhibitors	$\frac{1}{2} \times 4 = 2$	CO1 & CO2

Jharkhand University of Technology, Ranchi**B.Tech. 1st Semester Examination, 2024 (NEP)****Subject : Engineering Chemistry****Subject Code : BSC02****Time Allowed : 3 Hours****Full Marks : 70**

Candidates are requested to give the answers in their own words as far as practicable.

The figure in the right-hand margin indicate marks.

Answer any five questions in which Question No. 1 is compulsory.

1. Select & write the correct answer:

2×7=14

- (i) Electrochemical corrosion takes place on
(a) Anodic area (b) Cathodic area
(c) Near cathode (d) Near anode
- (ii) Anodic coating protects underlying metal
(a) due to higher oxidation potential. (b) due to noble character.
(c) due to higher reduction potential. (d) due to low reduction potential.
- (iii) An example of secondary battery cell is
(a) Edison alkali cell (b) Daniel cell
(c) Leclanche cell (d) Bunsen cell
- (iv) The storage battery generally used in electric power station is
(a) Lead acid battery (b) Zinc carbon battery
(c) Nickel cadmium battery (d) None of these
- (v) For nanoparticles with size below 2 nm, the localized carriers are in the _____ band.
(a) 5d (b) 3d
(c) 4d (d) None of these
- (vi) Presence of functional groups in a compound can established by using
(a) IR spectroscopy (b) UV spectroscopy
(c) Chromatography (d) None of these
- (vii) Which of the following is least polar for chromatographic separation?
(a) Benzene (b) Water (H₂O)
(c) Di-ethyl ether (d) Ethyl alcohol

2. (a) Discuss the role of following materials in engineering fields: 3½x2
(i) Ceramic materials
(ii) Composite materials
(b) Write the principle of infra-red spectroscopy. Discuss various types of vibration in IR spectroscopy? 7
3. (a) Define corrosion. Discuss about various factors affecting corrosion.
(b) Discuss about cathodic & anodic protection method of corrosion. 7+7
4. (a) What do you mean by primary & secondary battery? Differentiate between them with suitable examples.
(b) Discuss principle, construction, working & application of lithium ion battery. 7+7
5. (a) What are nanomaterials? Explain the optical & electrical properties of nanomaterials with suitable examples.
(b) What is sol-gel process for the synthesis of nanomaterials. Write basic steps involved in this process. 7+7
6. Explain the following:
(a) Emerging trends of analytical techniques for engineering
(b) Impact of corrosion on engineering materials 7+7
7. Write notes on the following: 3½x4=14
(a) Thermogravimetric Analysis (TGA)
(b) Corrosion inhibitors
(c) Green chemistry
(d) Super capacitor
-

Jharkhand University of Technology, Ranchi**B.Tech. 1st Semester Examination, 2023 (NEP)****Subject : Engineering Chemistry****Subject Code : BSC02****Time Allowed : 3 Hours****Full Marks : 70***Candidates are requested to give the answers in their own words as far as practicable.**Answer any five questions in which Question No. 1 is compulsory.**All questions are of equal value.***1. Answer the following questions :****2×7=14**

- (i) Which of the following wavelength fall in IR spectrum?
- (a) 2000 nm (b) 110 nm
(c) 800 nm (d) 10 cm
- (ii) Identification of an organic compound that can established by using
- (a) IR spectroscopy (b) UV spectroscopy
(c) Any of these (d) None of these
- (iii) The metal at the top of electrochemical series is
- (a) More active (b) Less active
(c) Most stable (d) More noble
- (iv) Difficult to monitor and very dangerous form of corrosion—
- (a) Pitting (b) Galvanic
(c) Crevice (d) Stress
- (v) The nanomaterials have the following characteristics—
Which do not exist in the corresponding bulk materials?
- (a) High surface energy (b) Greater imperfection
(c) Smaller number of surface atom (d) None of these
- (vi) Which of the following nanomaterials is utilized as a solid-fuel in rocket propulsion?
- (a) Nano aluminium (b) Nano zirconium
(c) Nano silicate (d) All of these
- (vii) The capacity of a cell is measured in
- (a) Ampere - hour (b) Amperes
(c) Watts (d) Watt - hours

2. (a) Discuss about following:
- (i) Quality control in engineering context
 - (ii) Role of refractories in engineering
- (b) Write the principle of UV spectroscopy. Discuss various electronic transitions. (3.5×2)+7
3. (a) State and explain electrochemical theory of corrosion. 7+7
- (b) Discuss about various corrosion prevention techniques.
4. (a) Define battery. Differentiate between electrolytic cell and electrochemical cell. 7+7
- (b) Explain principle, working and application of fuel cells.
5. (a) Define nanomaterials. Explain the mechanical and magnetic properties of nanomaterials. 7+7
- (b) Explain various importance of nanomaterials.
6. (a) Explain top-down and bottom-up approach of nanomaterials. 7+7
- (b) Discuss about green hydrogen technology.
7. Write notes on following: 3.5×4
- (a) chromophores
 - (b) auxochromes
 - (c) HPLC
 - (d) Galvanic corrosion

B. I. T. SINDRI, DHANBAD
MECHANICAL ENGINEERING DEPARTMENT
B.TECH FIRST SEMESTER
FIRST MID SEMESTER EXAMINATION DEC-2024

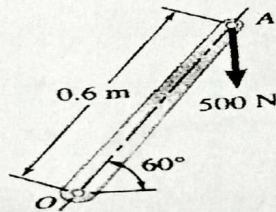
Branch: Mech, civil, Metal, Chemical, Prod, Mining
Subject: ESE02

Maximum Marks: 10
Time : 90 Minute

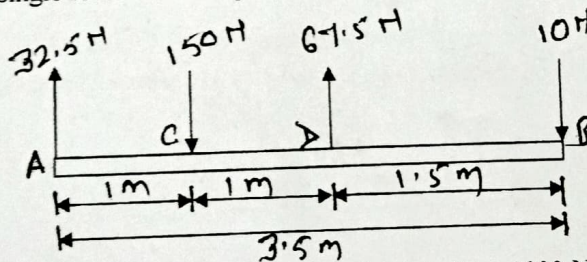
Instructions to Candidate

- Answer any five questions.
- All questions are of equal value.
- Missing data, if any, may be assumed suitably.

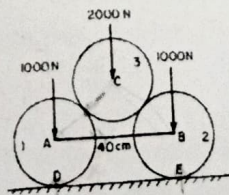
1. A 500-N vertical force is applied to the end of a lever, which is attached to a shaft at O. Determine (a) the moment of the 500-N force about O; (b) the horizontal force applied at A that creates the same moment about O; (c) the smallest force applied at A that creates the same moment about O; (d) how far from the shaft a 1250-N vertical force must act to create the same moment about O.



2. A system of parallel forces is acting on a rigid bar as shown in Fig. Reduce this system to:
- A single force
 - A single force and a couple at A
 - A single force and a couple at B.

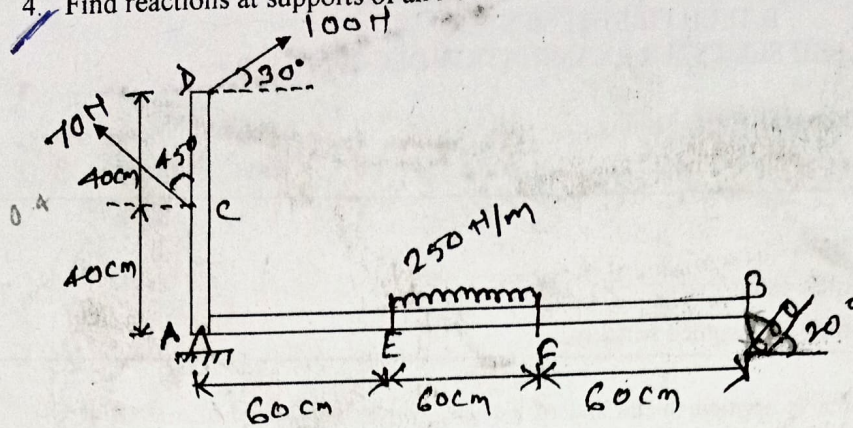


3. Two smooth circular cylinders, each of weight $W = 1000$ N and radius 15 cm, are connected at their centers by a string AB of length = 40 cm and rest upon a horizontal plane, supporting above them a third cylinder of weight = 2000 N and radius 15 cm as shown in Fig. Find the force S in the string AB and the pressure produced on the floor at the points of contact D and E.

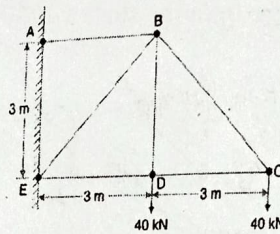


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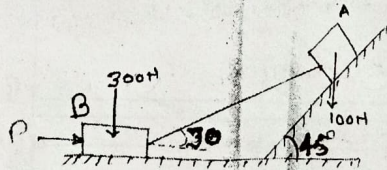
4. Find reactions at supports of an I-bent shown in fig.



5. Find the forces in all the members of the truss shown in fig. Tabulate the result.



6. A block A weighing 100 N rests on a rough inclined plane whose inclination to the horizontal is 45° . This block is connected to another block B weighing 300 N resting on a rough horizontal plane, by a weightless rigid bar inclined at 30° to the horizontal. Find the horizontal force required to be applied to the block B to just move the block A in upward direction. Assume angle of limiting friction as 15° at all surfaces where there is a sliding.



Explain any two:

- Moment of Couple
- Support reaction
- Angle of repose
- Free body diagram

Jharkhand university of Technology, Ranchi
2nd Mid Semester Exam - 2024 (1st Sem)

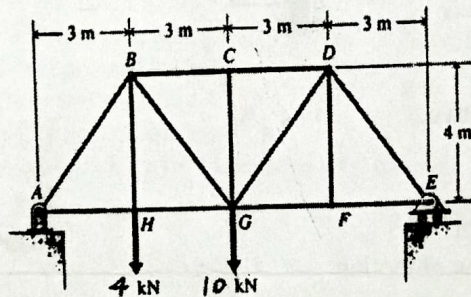
Subject: Engineering Mechanics
Time : 1.5 Hours

Branch: Mech/Civil/Metal/Chem/Prod/Min
Full marks: 10

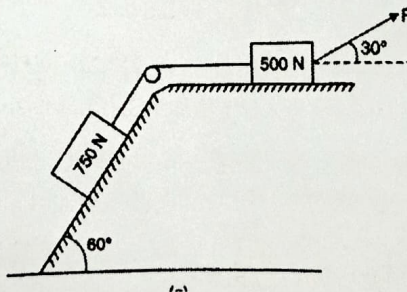
General Instructions:

- i. Write your name and Registration. no. clearly in question paper.
- ii. Answer any five questions.
- iii. Each question carry equal marks.(2m)

- (1) Determine the force carried in truss members CD, DG, and FG using method of section. [CO2]

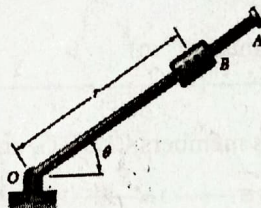


- (2) what is the value of P in the system shown in figure below to cause the motion of 500N block to the right side? Assume pulley is smooth and coefficient of friction between contact surfaces is 0.20. [CO3]



- (3) A stone is dropped from a balloon going up with a uniform velocity of 5 m/s. If the balloon was 60 m high when the stone was dropped, find its height when the stone hits the ground. Take $g = 10 \text{ m/s}^2$. [CO4]
- (4) A pilot flying his bomber at a height of 2000m with a uniform horizontal velocity of 600kmph wants to strike a target. At what distance from the target he should release the bomb? [CO4]

- (5) A car accelerates from rest at a constant rate α for some time after which it decelerates at constant rate β to come to rest. If the total time elapsed is t . Find the maximum velocity acquired by the car and total distance travelled during the time t . [CO4]
- (6) Rotation of the arm about O is defined by $\theta = 0.15t^2$ where θ is in radians and t in seconds. Collar B slides along the arm such that $r = 0.9 - 0.12t^2$ where r is in meters. After the arm has rotated through 30° , determine (a) the total velocity of the collar (b) the total acceleration of the collar. [CO4]



7) write short notes on any two.

- (a) limiting friction
- (b) relative velocity
- (c) n-t coordinate system
- (d) Newton's second law of motion

[CO3,CO4]

BIT SINDRI DHANBAD (JHARKHAND) – 828 123
Mid Semester Exam - II (MECH & PRODUCTION - 2024)
1st Semester, Section - A & B
SUBJECT: ENGINEERING MECHANICS

Time Allowed: 90 Minutes

Full Marks: 20

INSTRUCTIONS

1. There are 06 questions answer any 04 questions. Each question carries equal marks.
2. Notations are their usual meaning.
3. Assume suitable data for any missing data.

-
- Q.1** a) Write the scalar components of the acceleration and also show its directions, if a particle moving with velocity, v on a curved path when radius of curvature is ρ . [3]
- b) The position of a particle which moves along a straight line is defined by the relation, $x = t^3 - 6t^2 - 15t + 40$, where x is expressed in meters and t in seconds. Determine:
- i) the time at which the velocity will be zero,
 - ii) the distance traveled by the particle at that time,
 - iii) the distance traveled by the particle from $t = 4$ second to $t = 6$ second. [3]
- Q.2** a) Write the expression for distance travelled in n^{th} second of a body is having linear motion. [2]
- b) A ball is tossed with a velocity of 10 m/s directly vertically upward from a window located 20 m above the ground. Knowing that the acceleration of the ball is constant and equal to 9.81 m/s^2 downward. Determine:
- i) the velocity v and elevation y of the ball above the ground at any time t ,
 - ii) the highest elevation reached by the ball and corresponding value of t ,
 - iii) the time when the ball will hit the ground and the corresponding velocity.
- Draw the $v - t$ and $y - t$ curves. [3]
- Q.3** a) Define linear momentum and linear impulse of a particle of mass, m acted upon by a force, F . [2]
- b) A 10 kg package drops from a downhill into a 25 kg cart with a velocity of 3 m/s. Knowing that the cart is initially at rest and roll freely. Determine:
- i) the final velocity of the cart,
 - ii) the impulse exerted by the cart on the package,
 - iii) the fraction of the initial energy lost in the impact. [3]
- Q.4** Car A and Car B are moving on a road with different velocities is shown in Fig. 1. The velocity and tangential acceleration of car A are $v_A = 10 \text{ m/s}$ and $(a_t)_A = 5 \text{ m/s}^2$ respectively whereas velocity and acceleration of car B are given by, $v_B = 18.5 \text{ m/s}$ and $a_B = 2.0 \text{ m/s}^2$. Find: a) $v_{A/B}$ and b) $a_{A/B}$ [5]

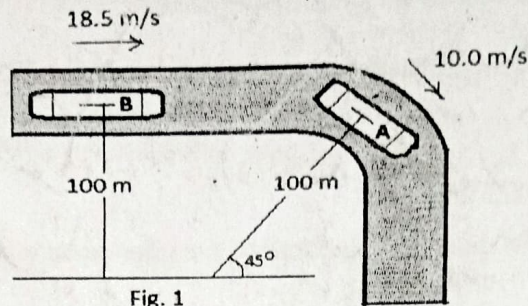


Fig. 1

- Q.5** A 1000 kg car starts from rest at point 1 and moves without friction down the track shown in Fig. 2. Determine:
- the force exerted by the track on the car at point 2, where the radius of curvature of the track is 6.0 m .
 - the minimum safe value of the radius of curvature at point 3. [5]

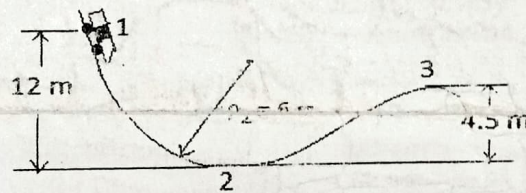


Fig. 2

- Q.6** Two blocks A and B of weights 1.0 kN and 2.0 kN respectively are in equilibrium positions as shown in Fig. 3. If the coefficient of friction between the two blocks as well as the block B and the floor is 0.3 , find the force, P required to move the block B. [5]

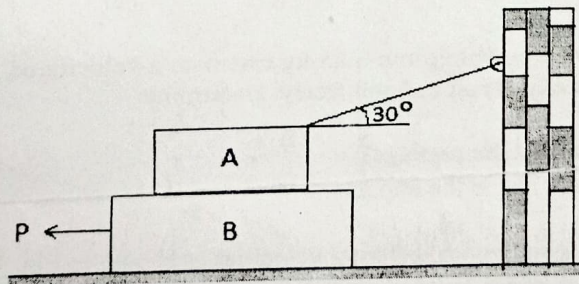


Fig. 3

Jharkhand University of Technology, Ranchi

B.Tech. 1st Semester Examination, 2023 (NEP)

Subject : Engineering Mechanics

Subject Code : ESEM1

Time Allowed : 3 Hours

Full Marks : 70

Candidates are required to give the answers in their own words as far as practicable.

Answer any five Questions in which Question No.1 is compulsory.

All questions are of equal value.

2×7=14

1. Choose the correct alternative:

(i) What is the dot product of two vectors which are having a magnitude equal to unity and are making an angle of 45° ?

(a) - 0.707

(b) 0.707

(c) - 1.414

(d) 1.414

(ii) The moment is the cross product of which two vectors?

(a) Radius and Force vectors

(b) Force and Radius vectors

(c) Radius and Force scalars

(d) Force and Radius scalars

(iii) The possible loading in various members of framed structures are

(a) compression or tension

(b) buckling or shear

(c) shear or tension

(d) all of these

(iv) Frictional force encountered after commencement of motion is called

(a) Limiting friction

(b) Kinematic friction

(c) Frictional resistance

(d) Dynamic friction

(v) The radial component of velocity for a particle moving in circular path is _____.

(a) constant

(b) radius itself

(c) variable

(d) zero

(vi) A bus travels with a speed of 15 m/s when accelerated at 0.10 m/s^2 from its rest position. What is the distance travelled?

(a) 1000 m

(b) 1125 m

(c) 2250 m

(d) None of these

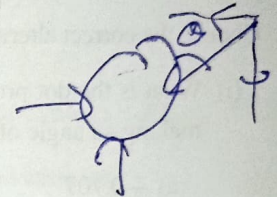
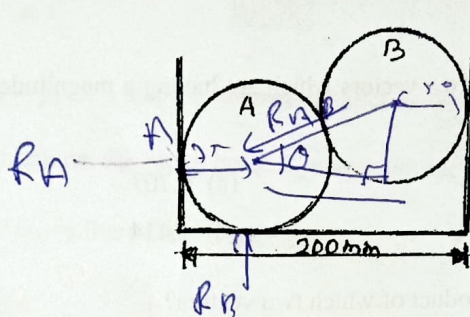
(vii) An object is released from a certain height above the ground. Just at time it touches ground it will possess _____.

- (a) Potential energy (b) Kinetic energy
(c) Heat energy (d) Chemical energy

2. (a) Explain moment of a force and moment of couple.

(b) Two spheres A and B of weight 1000 N and 750 N respectively are kept as shown in figure. Determine the reactions at all contact points. Radius of A is 75 mm and radius of B is 50 mm.

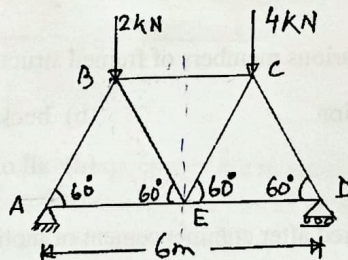
4+10



3. (a) What are the assumptions in analysis for plane truss?

(b) Find the forces in all the members of the truss shown below.

4+10



4. (a) Define coefficient of friction and limiting friction.

(b) A uniform ladder of 4 m length rests against a vertical wall with which it makes an angle of 45° . The coefficient of friction between ladder and wall is 0.4 and that between ladder and the floor is 0.5. If a man, whose weight is one-half of the ladder ascends it, how high will it be when the ladder slips?

4+10

5. (a) Derive the tangential and normal components of velocity and acceleration for a moving particle.

(b) The acceleration of a particle is given by $a = t^3 - 3t^2 + 5$, where a is in m/s^2 and t is in second. At $t = 1$ s, the displacement and velocity are 8.5 m and 6.25 m/s respectively. Calculate the displacement and velocity at $t = 2$ sec.

7+7

(3)

6. (a) Explain work, energy and power.

(b) A hammer of mass 0.5 kg hits a nail of 25 g with a velocity of 5 m/s and drives it into a fixed wooden block by 25 mm. Find the resistance offered by the block. 6+8

$$\begin{aligned} W &= \frac{F \times D}{F \times 25} = \frac{1}{2} (m) \times 2 \\ &= K_f - K_i \\ &= \end{aligned}$$

7. Write short notes on any two of the following:

(a) Belt friction

(b) Impulse momentum principle

(c) Equivalent system of forces

Jharkhand University of Technology, Ranchi

B.Tech. 1st Semester Examination, 2024 (NEP)

Subject : Engineering Mechanics

Subject Code : ESEM1

Time Allowed : 3 Hours

Full Marks : 70

Candidates are required to give the answers in their own words as far as practicable.

Answer any five Questions in which Question No.1 is compulsory.

The figures in the right-hand margin indicate the marks.

1. Choose the correct alternative:

2×7=14

(i) If two equal vector forces are mutually perpendicular, then the resultant force is acting at which angle as compared to one of the vector is

- | | |
|----------------|-----------------|
| (a) 0° | (b) 45° |
| (c) 90° | (d) 180° |

(ii) The tendency of rotation of the body along any axis is also called _____.

- | | |
|----------------------|-----------------------|
| (a) Moment of couple | (b) Moment of inertia |
| (c) Force | (d) Torque |

(iii) The product of either force of couple with the arm of the couple is called

- | | |
|-----------------------|---------------------------|
| (a) Resultant couple. | (b) Moment of the forces. |
| (c) Resulting couple. | (d) Moment of the couple. |

(iv) The coefficient of friction depends on

- | | |
|-----------------------|--------------------------|
| (a) area of contact | (b) shape of surfaces |
| (c) nature of surface | (d) strength of surfaces |

(v) In curvilinear motion, what is the normal component of acceleration?

- | | |
|----------------------|------------------------|
| (a) $\frac{a^2}{t}$ | (b) $\frac{v^2}{\rho}$ |
| (c) $\frac{\rho}{v}$ | (d) $\frac{dv}{dt}$ |

EMI

(2)

(vi) A car moves in a straight line that for a short time its velocity is defined by $V = (3t^2 + 2t)$ m/s, where 't' is in seconds, determine its position when $t = 3$ sec.

(a) 54 m

(b) 36 m

(c) 18 m

(d) 9 m

(vii) Which of the following is represented by the area under force-displacement diagram?

(a) Impulse

(b) Momentum

(c) Power

(d) Work done

2. (a) Define moment and couple moment. Show that moment of a couple is a free vector.

(b) Find the support reactions at A and P if reaction at B is 60 kN for the beam loaded as shown in figure 1. 4+10

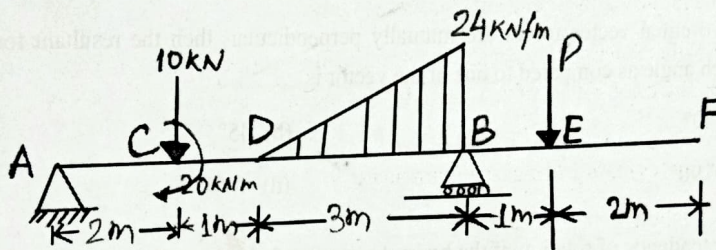
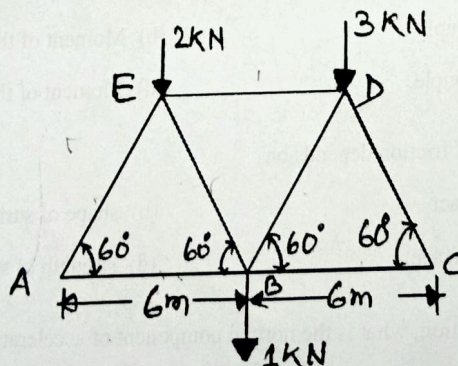


Fig-1

3. (a) What are the assumptions in analysis for plane truss?

(b) Find the forces in all the members of the truss shown below. 4+10



4. (a) Derive the tangential and normal components of velocity and acceleration for a moving particle.

(b) A particle moves along a curved path defined by $r = 5\theta$ and $\theta = t^2/3$, where r is in meters and t is in seconds. Calculate velocity and acceleration of the particle when $\theta = \pi/2$. 7+7

(3)

EMI

5. (a) Define angle of friction and angle of repose.

(b) Two blocks of weight 50N and 200N and connected by a cord rest on two inclined planes as shown in figure 2. Determine the maximum tension in the cord when limiting friction conditions develop for both the blocks. (Take $\mu = 0.25$). 4+10

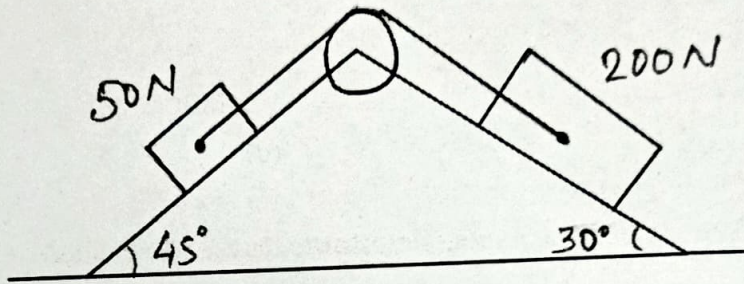


Fig-2

6. (a) Define impulse, linear and angular momentum.

(b) Two balls of masses 2 kg and 3 kg are moving with velocities 2 m/s and 3 m/s towards each other. If the coefficient of restitution is 0.5, find the velocity of the two balls after impact. 6+8

7×2=14

7. Write short notes on *any two* of the following:

(a) Belt friction

(b) Work energy principle

(c) Equivalent system of forces
